

Weekly Report

CARLA Fleet Control and Ground-Station Integration

Qi LI

July 24, 2026

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1	Work summary	

This week I completed:

- Completed the versioned TCP ground-station bridge, server, command-line interface, and read-only terminal dashboard. And a three-vehicle CARLA fleet scenario is presented here with one leader, two followers, predecessor V2V exchange, and recorded spacing, speed, steering, and fleet-health diagnostics.
- Kept the current result focused on platform integration. The observer theory and the SDCS small-map path-following result are deferred to the next weekly report.

2 Fleet control

2.1 Fleet composition and runtime flow

The CARLA experiment contains three vehicles: one leader and two followers. Figure 1 shows the repeated local structure and data flow. Each vehicle keeps its own local estimate and actuator path; only typed V2V predecessor-state messages are exchanged between vehicles.

The formation becomes active only after each follower receives a current predecessor state. A missing, stale, invalid, or out-of-order message triggers the fleet safety response;

The local runtime remains the only component that writes actuator commands.

2.2 Follower controller and distributed observer

The leader uses the normal *pathplanner* and *ego controller*. Each follower uses the **ControllerFleet**: it

places a virtual target behind its direct predecessor and tracks that target. With follower speed v_i , predecessor pose (x_p, y_p, θ_p) , and follower pose (x_i, y_i, θ_i) , first compute the target distance d^* and speed v^* :

$$d^* = d_0 + h \max(0, v_i)$$

$$v^* = \text{sat}_{[0, v_{\max}]}(v_p + k_d(\|p_p - p_i\| - d^*))$$

then compute the virtual target pose (x_v, y_v, ψ_v) and steering command δ by:

$$p_v = p_p - d^* [\cos \theta_p, \sin \theta_p]^\top$$

$$\psi_v = \text{atan2}(y_v - y_i, x_v - x_i)$$

$$a = \text{sat}_{[-a_{\max}, a_{\max}]}(k_v * (v^* - v_i))$$

$$\delta = \text{sat}_{[-\delta_{\max}, \delta_{\max}]}(k_\psi \text{wrap}(\psi_v - \theta_i))$$

Here $d_0 = 5$ m and $h = 0.5$ s. The speed error produces bounded throttle, and δ is the bounded steering command.

DistributedObserverLuenberger is an advisory prediction-correction prototype. It combines the local estimate with validated peer snapshots to form a collective estimate for diagnostics. It does not provide a controller reference or actuator command, so this report does not claim a validated distributed-observer control result. Observer theory and validation remain the next-report task.

2.3 Fleet trajectory and spacing result

The recorded run executed 800 cycles per vehicle (about 40 s at 50 ms per cycle). Figure 2 shows the CARLA trajectories, physical predecessor spacing, vehicle speeds, and steering commands. After the initial formation transient, both predecessor spacings converge close to their speed-dependent targets while the three vehicles continue through the curved route.

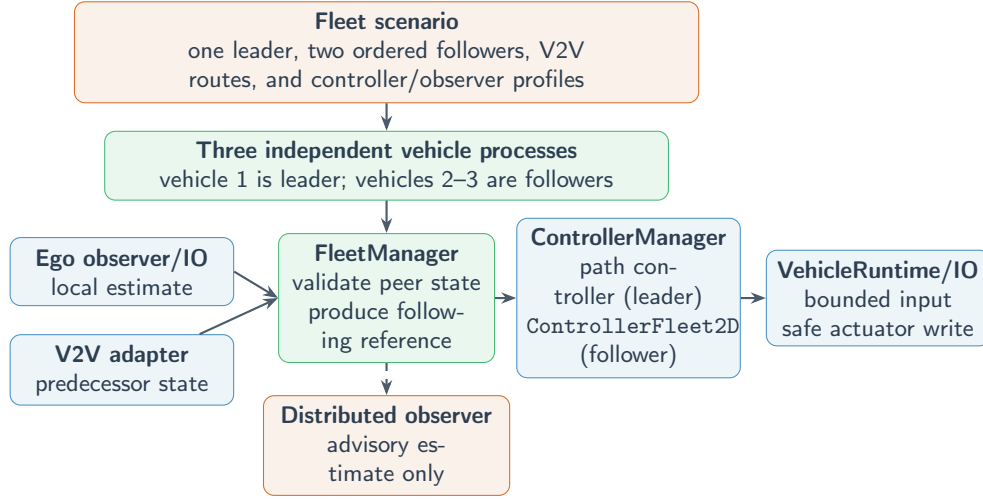


Figure 1: Fleet composition and runtime flow. Each vehicle process repeats the same local structure; only typed V2V predecessor-state messages cross between processes. The distributed observer remains advisory and has no actuation path.

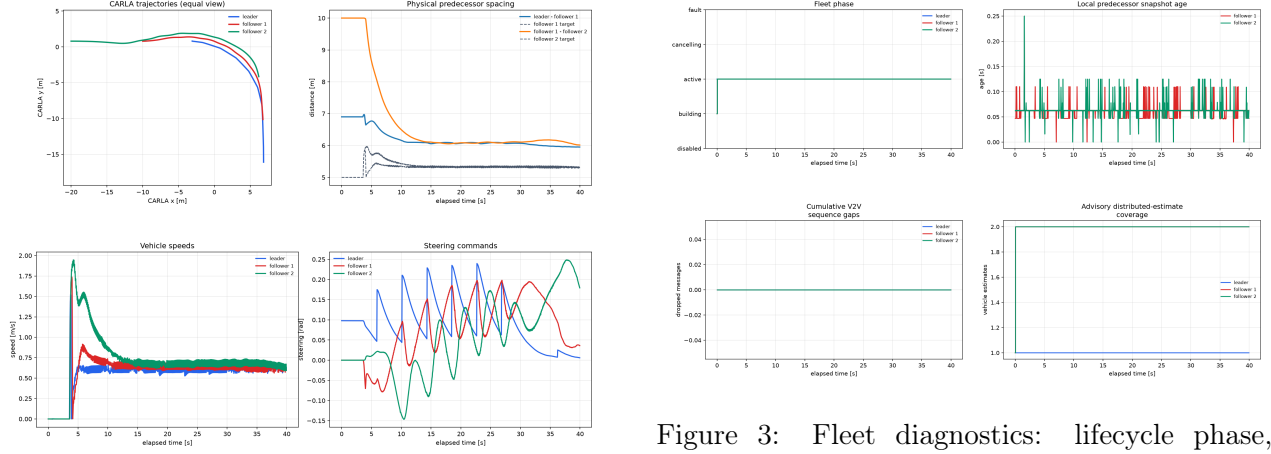


Figure 2: Three-vehicle CARLA fleet result: trajectories, predecessor spacing, speed, and steering command.

Figure 3 shows that the formation moves from building to active at the start of the run and remains active. The V2V sequence-gap counter remains zero in this local CARLA run, and both followers keep advisory distributed estimates for the other two vehicles.

Figure 3: Fleet diagnostics: lifecycle phase, predecessor-message age, V2V sequence gaps, and distributed-estimate coverage.

Check	Recorded result
Fleet composition	Three processes: one leader and two ordered followers.
Runtime duration	800 cycles per vehicle, corresponding to approximately 40 s.
Follower control	Both followers enter the active phase and use the fleet 2D controller with non-zero throttle and steering.
Safety spacing	The integration check requires a physical gap of at least 4 m and bounds steady-state error relative to the speed-dependent spacing reference.
V2V health	Predecessor freshness and sequence-gap counters are recorded; this local run shows no sequence gaps.
Distributed observer	The leader keeps one local estimate and each follower maintains advisory coverage for all three fleet vehicles.

Table 1: Recorded CARLA fleet-control evidence.

3 Ground-station integration

3.1 Design and safety boundary

The ground station is an external operator process, not part of the vehicle runtime.

It exchanges bounded, versioned MessagePack frames over TCP. The vehicle-side bridge runs its socket work in a background thread, but that thread never calls the runtime directly. Instead, the runtime facade drains queued typed commands at its normal step boundary, returns acknowledgements, and publishes monitoring snapshots.

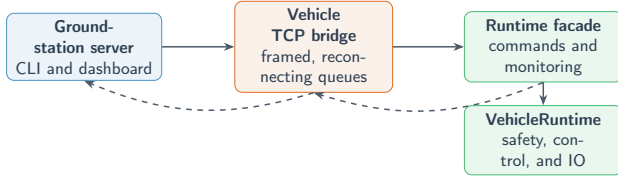


Figure 4: Ground-station communication boundary. Transport queues data but does not bypass vehicle safety or direct IO ownership.

3.2 Difference from the original ground station

The structure of the ground station has changed significantly.

The original ground station was a GUI-centred application with one TCP listener per vehicle. The current ground station is a protocol-centred operator service: all vehicles connect to one configured listener and identify themselves with a versioned registration frame.

This makes the server independent of a fixed vehicle count and allows it to represent connection, stale, and reconnection states explicitly.

Aspect	Original ground station	Current ground station
Network identity	One listener for each vehicle; identity inferred from the port.	One listener; a vehicle sends a versioned <code>REGISTER</code> frame containing its ID.
Scheme	GUI, connection handling, and vehicle-specific workflows were closely coupled.	CLI/dashboard, server routing, TCP bridge, runtime facade, and vehicle runtime have separate responsibilities.
Command path	Commands were carried in application-specific messages to a vehicle connection.	Typed <code>VehicleCommand</code> values are validated at the runtime step boundary and return a correlated acknowledgement.
Safety	Ground-station features were integrated with the legacy vehicle application.	Transport cannot call the runtime or write actuators; <code>VehicleRuntime</code> remains the sole safety and IO owner.
Operator evidence	GUI-oriented status and feature views.	Uniform monitoring snapshots expose runtime/fleet phase, estimates, V2V health, peer age, stale/disconnected state, and command outcome.
Compatibility	Legacy port and message conventions.	Deliberately a new bounded MessagePack protocol; no per-vehicle-listener compatibility bridge is retained.

Table 2: Architectural comparison of the original and current ground stations.

3.3 Live operator view

Figure 5 is a live snapshot taken during the three-vehicle CARLA fleet experiment. The upper part of the screenshot shows the three vehicles in the CARLA map. The terminal ground station below it is the operator view of the same experiment.

The **Vehicle status** panel reports that all three vehicles are connected and **RUNNING**. For each vehicle, it displays the fleet phase and role (leader or follower), estimated state, current reference, estimate validity, peer age, V2V received/loss/dropped counters, update rate, and monitoring age. The **Commands and acknowledgements** panel shows the command sent to the selected vehicle and the returned `COMMAND_ACK`; for example, the screenshot records start and velocity commands that were applied. The bottom **Command input** panel is the operator entry point for fleet-safe commands such as `start`, `stop`, `emergency-stop`, and `set-velocity`.

3.4 Implemented capabilities

- **Framed protocol:** a length-prefixed, versioned MessagePack protocol supports registration, monitoring, command, acknowledgement, and error frames. The decoder accepts fragmented TCP reads and rejects invalid or oversized frames.
- **Reliable operator connection:** the vehicle bridge registers with the server, reconnects after a listener restart, retains only the latest monitoring snapshot, and bounds command and acknowledgement queues.
- **Safe command path:** CLI requests are converted to typed `VehicleCommand` instances. The runtime remains responsible for applying or rejecting commands and returns the resulting state to the operator.
- **Monitoring dashboard:** the terminal display shows connection state, runtime state, estimate, control reference, V2V counters, fleet peer freshness, command outcome, and stale/disconnected status.

3.5 Verification result

The ground-station protocol, bridge, runtime facade, server, and dashboard were verified by the unit test suite. The 28 checks passed, covering frame fragmentation and validation, command conversion, command-snapshot-acknowledgement flow, duplicate ID rejection, stale and disconnected dashboard rows, listener restart/reconnection, bounded queues, and manual-input coalescing.

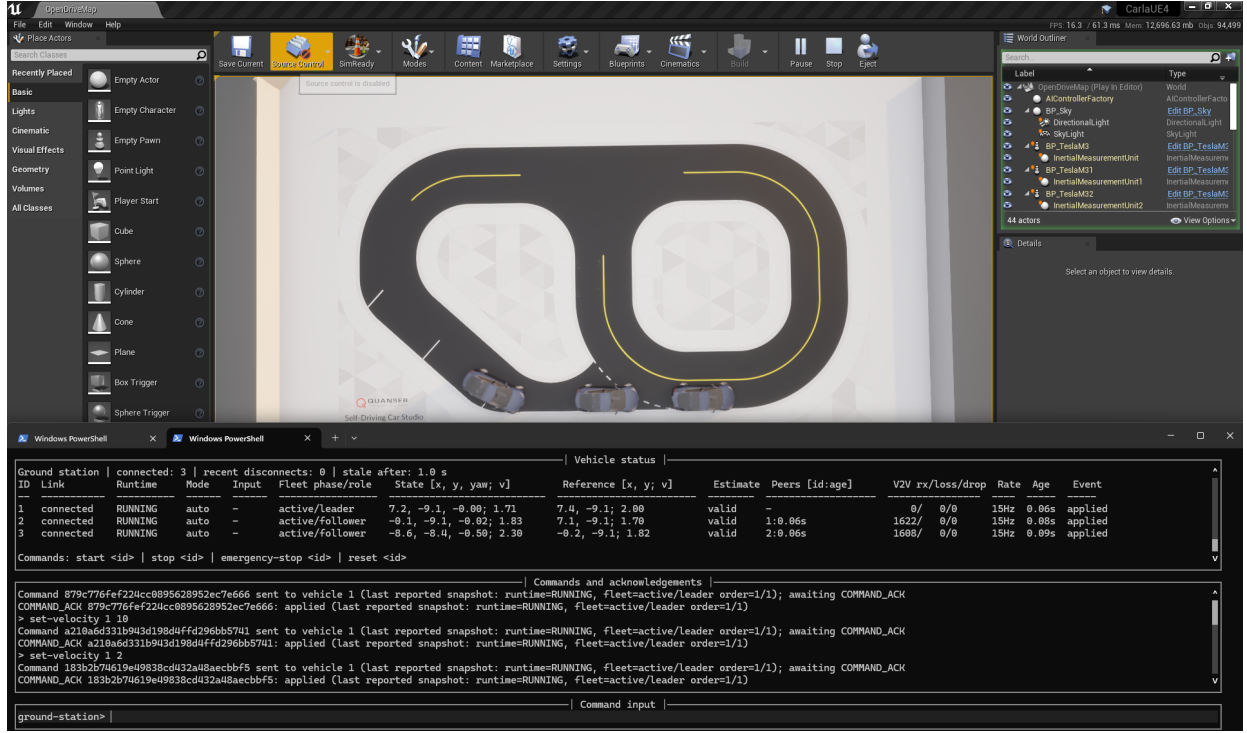


Figure 5: Ground-station screenshot during the three-vehicle CARLA fleet run. The CARLA scene is shown above; the operator dashboard, command log, and command input are shown below.

Area	Verified behaviour
Protocol	Incremental frame decoding, maximum frame size, and protocol-version validation.
Command flow	Typed commands arrive at the vehicle bridge, are drained by the facade, and return an acknowledgement.
Connection lifecycle	Vehicle registration, duplicate live-ID protection, stop handling, and reconnect after a server restart.
Dashboard	Latest monitoring data, stale sessions, disconnected sessions, fleet fault reason, and missing-peer display.
Queue policy	Bounded normal command queue and newest-value coalescing for high-rate manual input.

Table 3: Ground-station verification coverage.

4 Conclusion and next step

The refactored platform now has a three-vehicle CARLA fleet result and an operator-facing ground-station boundary. The fleet artifacts show formation activation, predecessor spacing, follower control, and V2V health; the ground station provides controlled commands and readable monitoring without giving the transport layer direct control authority.

The next weekly report will add the theoretical observer result and the SDCS small-map path-following result. Those results will build on this verified fleet, CARLA, V2V, and ground-station foundation.